

Additional Features Analysis (Beyond Specification)

This document lists the fully implemented features in the `EnergyV1` project that were **not** requested in the original `Website_Fitur.xlsx` specification but exist in the current codebase. These represent significant added value and functionality.

Executive Summary

- **Total Additional Features Found:** 9
- **Overall Impact:** Highly positive. These features elevate the system from a basic product catalog to a comprehensive e-commerce and lead-generation platform with a robust enterprise-grade CMS.

Summary Table

Feature	Current Status	Description	Added Value
1. Project / Portfolio Showcase	Complete	Dedicated pages for past installations (/proyek).	Builds trust and showcases technical capability.
2. Customer Testimonials	Complete	Testimonial system with ratings and project linking.	Provides social proof to potential buyers.
3. Wishlist System	Complete	Users can save products to a wishlist (/wishlist).	Improves user retention and future conversions.
4. FAQ Section	Complete	Dedicated Frequently Asked Questions page (/faq).	Reduces repetitive inquiries to customer service.
5. Comprehensive Full CMS	Complete	Full admin panel beyond just the requested "News".	Total control over site content, products, and leads.
6. Activity & Audit Logs	Complete	System tracks all admin actions and entity revisions.	Essential for team accountability and mistake recovery.
7. Product Analytics / Tracking	Complete	Tracks views, quote requests, and interactions.	Provides data-driven insights into popular products.
8. Clearance Category	Complete	Special routing and tagging for clearance items.	Helps clear old stock efficiently.
9. Role-Based Auth	Complete	Admin system with Owner, Admin, Editor roles.	Secure, multi-user environment management.

Feature Details

1. Project / Portfolio Showcase

Current Status:

Complete

- **Description:** The platform includes a comprehensive "Proyek" (Projects) system where admins can publish case studies of past installations (On-Grid, Off-Grid, Hybrid).
- **Current Implementation:** Features rich descriptions, galleries, system specs, and SEO data.
- **Evidence:** `src/app/proyek` , `src/app/admin/projects` , and `Project` database model.
- **Why it matters:** Significantly boosts company credibility by proving past success.

2. Customer Testimonials

Current Status:  Complete

- **Description:** A system to collect, manage, and display customer reviews and ratings.
- **Current Implementation:** Reviews can be linked to specific products or projects and displayed throughout the site.
- **Evidence:** `src/app/testimoni` , `src/app/admin/testimonials` , and `Testimonial` database model.
- **Why it matters:** Crucial for enterprise B2B sales and high-ticket B2C purchases.

3. Wishlist System

Current Status:  Complete

- **Description:** Users can "favorite" or save products they are interested in but not yet ready to buy.
- **Current Implementation:** Implemented via `WishlistToggleButton` on product cards and a dedicated `/wishlist` page.
- **Evidence:** `src/app/wishlist` , `src/components/product/WishlistToggleButton.tsx` .
- **Why it matters:** Reduces friction for returning users who are in the consideration phase.

4. FAQ Section

Current Status:  Complete

- **Description:** A dedicated section to answer common customer queries regarding shipping, installation, and warranties.
- **Current Implementation:** Backed by the database allowing admins to reorder and manage active questions.
- **Evidence:** `src/app/faq` , `src/app/admin/faq` , and `Faq` database model.
- **Why it matters:** Enhances customer support and self-service capabilities.

5. Comprehensive Full CMS

Current Status:  Complete

- **Description:** The Excel file only asked for "Akses Admin untuk Portal Berita". The project actually contains a full-blown Enterprise CMS.
- **Current Implementation:** Admins can manage Products, Categories, Homepage layouts, Settings, Quotes (CRM), Banners, Projects, and static pages dynamically.
- **Evidence:** `src/app/admin/*` routing, `SiteSetting` and `HomepageSection` models.
- **Why it matters:** Zero-code maintenance for business owners going forward.

6. Activity & Audit Logs

Current Status:  Complete

- **Description:** The backend records who made what changes to the site and saves snapshots of previous data.

- **Current Implementation:** Tracks creations, updates, deletions, and stores JSON snapshots of entities.
- **Evidence:** `ActivityLog` and `Revision` models in `prisma/schema.prisma` .
- **Why it matters:** Protects against accidental data loss or unauthorized changes by team members.

7. Product Analytics / Tracking

Current Status:  Complete

- **Description:** Internal telemetry for tracking which products are getting the most views and quote requests.
- **Current Implementation:** Automatically logs events when users view products, add to compare, or request quotes.
- **Evidence:** `ProductViewTracker.tsx` and `ProductEvent` database model.
- **Why it matters:** Gives the marketing team actionable data on user interest.

8. Clearance Category

Current Status:  Complete

- **Description:** A dedicated promotional section for clearing out old or "siswa proyek" stock.
- **Current Implementation:** Exists as a standalone route.
- **Evidence:** `src/app/barang-clearance/page.tsx` .
- **Why it matters:** Specific business logic adapted for inventory management.

9. Role-Based Auth

Current Status:  Complete

- **Description:** The admin panel isn't just a single password login; it supports multiple users with distinct permission levels.
- **Current Implementation:** NextAuth integration with roles: `owner` , `admin` , `editor` , and `viewer` .
- **Evidence:** `AdminUser` model and `src/app/api/auth/[...nextauth]` .
- **Why it matters:** Scales securely as the internal team grows.